

AYUSH KUMAR

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Summary

Curious and highly self-driven 3rd-year B.Tech CSE student with a strong interest in Machine Learning, emerging technologies and hands-on engineering. I enjoy exploring new tools, building practical systems, and solving problems through experimentation. Known for an optimistic mindset and rapid learning ability, I actively pursue projects that strengthen my technical depth and real-world development skills.

Education

Kalinga Institute of Industrial Technology (KIIT)

B.Tech in Computer Science and Engineering

I.G.N.S College, Kuchaikote

Senior Secondary (Physics, Chemistry, Mathematics)

Aug 2023 – Sept 2027

Bhubaneswar, India

June 2020 – June 2022

Bihar, India

Certifications

- Machine Learning — Kaggle (2025)
- Python — Kaggle (2025)
- Game AI and Reinforcement Learning — Kaggle (2025)

Projects

Connect 4 Web Game with AI | *Python, Django, JavaScript, NumPy, PyTorch*

Mar 2026

- Created **Minimax tree search** with **Alpha-Beta pruning**, cutting branching factor to $O(7^{2.5})$ and AI decision time to **<100ms** across ~2,400 nodes/turn.
- Applied **automated reasoning** via **PyTorch ValueNet** (3-layer MLP, 42-dim state vectors) for **neural-guided position evaluation** over hand-crafted heuristics.
- Optimized **move ordering** (DSA) to maximize Alpha-Beta cutoffs, reducing redundant subtree evaluations by **50%**.
- **Future:** Generalize into a **multi-game AI framework** (Chess, Checkers) via a **generic game-tree interface** reusing the same Minimax + ValueNet pipeline.

ScholarAgent AI Research Assistant | *Python, FastAPI, LangChain, FAISS, Gemini, Docker*

Dec 2025

- Built a multi-agent AI research assistant enabling semantic PDF search, contextual question answering, citation extraction, and downloadable research report generation using Retrieval-Augmented Generation (RAG).
- Designed modular retrieval, summary, citation, and report agents orchestrated through LangChain pipelines, reducing query-processing complexity and improving workflow separation.
- Implemented FAISS vector search with sentence-transformer embeddings to retrieve top-k relevant document chunks from 100+ page PDFs with low-latency semantic retrieval.
- Integrated FastAPI backend, Gemini LLM inference, and Streamlit frontend into a Dockerized full-stack AI system with REST APIs and containerized deployment support.

Sentiment Analysis Web App | *FastAPI, Scikit-learn, JavaScript*

Mar 2026

- Developed ML web app using TF-IDF + SVM achieving ~65% accuracy.
- Deployed REST API with real-time predictions and integrated responsive frontend.
- End-to-end deployment with full-stack integration and improved UI/UX.
- Handled multi-line input processing with aggregated sentiment output.
- **Future:** Add neutral class, live data integration, and visual dashboards.

DevArena — Real-Time Coding Duel Platform | *Java, PostgreSQL, Judge0 API*

Jan 2026

- Built backend for real-time 1v1 coding duels with match lifecycle and winner logic based on earliest correct submission.
- Integrated Judge0 API to execute code in 50+ languages securely.
- Designed PostgreSQL schema and scalable leaderboard system supporting 1,000+ users.
- Developed REST APIs for match creation, submissions, and user interactions.

Technical Skills

Programming Languages: Python, Java, HTML/CSS, JavaScript

Machine Learning / AI: Machine Learning, Neural Networks, Federated Learning, Deep Learning Fundamentals

Libraries / Frameworks: FastAPI, Streamlit, LangChain, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib

Tools / Technologies: Git, GitHub, Docker, REST APIs, JSON, FAISS, Postman, Jupyter Notebook

Databases: PostgreSQL